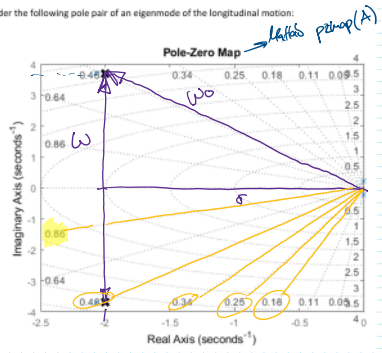
Exercise 10

Donnerstag, 22. Januar 2026 08:13

Exercise 10.1: Longitudinal Eigenmotion

Consider the following pole pair of an eigenmode of the longitudinal motion:



From the plot we read the following:

$\sigma \approx -2$

$\omega \approx \pm 3.7$

$\omega_0^2 = \sigma^2 + \omega^2 \Rightarrow \omega_0 = \sqrt{\sigma^2+\omega^2} = 4.2059 \frac{rad}{s}$;

$\omega_0 = 2\pi \cdot f \Rightarrow T = \frac{2\pi}{\omega_0} = 1.495$ s

$f = \frac{1}{T}$

↳ fast P → SP

a) Which eigenmotion is displayed here?

Short Periode Mode

b) Give an approximation for the derivatives, which define the eigenmode. You can neglect the influence of the pitch rate on the forces. Further simplifications have to be justified!

$\begin{bmatrix} \dot{\alpha} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} Z_\alpha & Z_q + 1 \\ M_\alpha & M_q \end{bmatrix} \begin{bmatrix} \alpha \\ q \end{bmatrix} + \begin{bmatrix} Z_\eta \\ M_\eta \end{bmatrix} \cdot \eta$

in the solution the simplification is mentioned but the derivation is conducted without the simplification

1) Solve determinante: $\det(sI - A) = 0$

$sI - A = \begin{bmatrix} s - Z_\alpha & -1 \\ -M_\alpha & s - M_q \end{bmatrix}$

$\det(sI - A) = (s - Z_\alpha)(s - M_q) - (-M_\alpha)(-1)$

$= s^2 - M_q \cdot s - Z_\alpha s + M_q Z_\alpha - M_\alpha$

2) Compare Coefficients: $= s^2 - (M_q + Z_\alpha)s + M_q Z_\alpha - M_\alpha \hat{=} s^2 - 2\sigma s + \omega_0^2 = 0$

($M_q + Z_\alpha = 2\sigma$; $M_q Z_\alpha - M_\alpha = \omega_0^2$)

Unit in A-Matrix per Entry: $\frac{1}{s}$ → $[1/s] \cdot [1/s] = \frac{1}{s^2}$

$\Rightarrow \omega_0^2 = M_q Z_\alpha - M_\alpha \cdot (Z_q + 1) = -3 \cdot (-1.14) + 15.34 = 18.76 \frac{1}{s^2}$ $\Rightarrow \omega_0 \sim 4$

$2\sigma = M_q + Z_\alpha = -3 - 1.14 = -4.14 \frac{1}{s}$; $\Rightarrow \sigma \sim -2$ } → about matches above

Evaluate with values from subtask c): $M_\alpha = -15.34$, $M_q = -3$, $Z_\alpha = -1.14$

c) Does the pole zero plot in 10.a) belong to the following linear state space model of a Boeing 777?

$\begin{pmatrix} \dot{v}^* \\ \dot{\alpha} \\ \dot{q} \end{pmatrix} = \begin{bmatrix} -0.01 & -0.49 & -0.046 & -0.001 \\ 0.11 & 0.0003 & 1.14 & 0.043 \\ -0.11 & -0.0003 & -1.14 & 0.957 \\ 0.1 & 0 & -15.34 & -3.00 \end{bmatrix} \begin{pmatrix} v^* \\ \alpha \\ q \end{pmatrix} + \begin{bmatrix} 0.0002 & 0.014 \\ -0.059 & 0 \\ 0.059 & 0 \\ 6.56 & 0.13 \end{bmatrix} \cdot \begin{pmatrix} \eta \\ \delta_T \end{pmatrix}$

$Z_q + 1 = 0.957 \rightarrow Z_q = -0.043$

Yes, see above in orange!

Exercise 10.2: Lateral Eigenmotion

The following linear model of the lateral motion is given:

$$\begin{pmatrix} \dot{r} \\ \dot{\beta} \\ \dot{p} \\ \dot{\Phi} \end{pmatrix} = \begin{bmatrix} -0.6 & 6.0 & -0.18 & 0 \\ -1 & -0.19 & 0 & 0.13 \\ 1.35 & -15.10 & -6.5 & 0 \\ 0.05 & 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} r \\ \beta \\ p \\ \Phi \end{pmatrix} + \begin{bmatrix} 0.0075 & -2.6 \\ 0 & 0.037 \\ -13.09 & 1.87 \\ 0 & 0 \end{bmatrix} \begin{pmatrix} \xi \\ \zeta \end{pmatrix}$$

a) Calculate the initial roll acceleration and the steady state roll rate for an aileron deflection of 1°.

→ The simplified State space model is:

$$\dot{p} = L_p \cdot p + L_\xi \cdot \xi$$

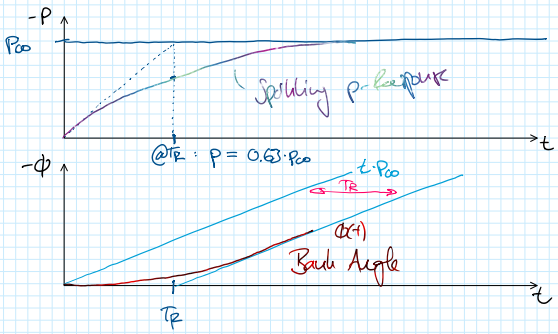
1) Initial Roll acceleration: $\xi = 1°$, $p = 0$

$$\dot{p}_0 = L_\xi \cdot \xi = -13.09 °/s^2$$

2) Steady State Roll Rate: $\xi = 1°$, $\dot{p} = 0$

$$0 = L_p \cdot p_\infty + L_\xi \cdot \xi \rightarrow p_\infty = -\frac{L_\xi}{L_p} \cdot \xi = -2.0138 °/s$$

b) Plot the step response of roll rate and bank angle for the given aileron deflection.



$$T_R = -\frac{1}{L_p} = 0.153s$$